MTH636 – Game Theory

Lecture 12: NE and the Indifference Principle

1 Minimax Strategies and Nash Equilibrium

For general two-player zero-sum games, finding max-min strategies can be framed as a Linear Programming problem (see Solan's book, Section 5.2.5). There is a profound connection between max-min strategies and Nash Equilibrium.

Theorem: Equivalence Theorem for Zero-Sum Games

For a two-player zero-sum game, a strategy profile (σ_1^*, σ_2^*) is a max-min strategy profile if and only if (σ_1^*, σ_2^*) is a Nash Equilibrium.

Proof Idea: If V is the value of the game, a max-min profile means $U(\sigma_1^*, \sigma_2) \geq V$ for all σ_2 and $U(\sigma_1, \sigma_2^*) \leq V$ for all σ_1. Setting $\sigma_2 = \sigma_2^*$ and $\sigma_1 = \sigma_1^*$ yields $U(\sigma_1^*, \sigma_2^*) = V$. Therefore:

$$U(\sigma_1^*, \sigma_2^*) = V \geq U(\sigma_1, \sigma_2^*) \quad \forall \sigma_1 \in \Sigma_1$$
$$U(\sigma_1^*, \sigma_2^*) = V \leq U(\sigma_1^*, \sigma_2) \quad \forall \sigma_2 \in \Sigma_2$$

This precisely satisfies the definition of a Nash Equilibrium. The proof in the other direction follows a similar bounding logic to show that the existence of an NE implies $\underline{V} = \overline{V} = V$.

Theorem: Security Level Theorem

Suppose σ^* is a Nash Equilibrium of any strategic form game (not necessarily zero-sum). Then every player achieves at least their max-min value (security level):

$$U_i(\sigma^*) \geq \underline{V}_i \quad \forall i \in N$$

2 Computing NE via Best Response

Let's compute the mixed Nash Equilibrium for a non-zero-sum game using best response correspondences.

Example: Battle of the Sexes

$P_1 \setminus P_2$	L (y)	R ($1-y$)
T (x)	(2,1)	(0,0)
B ($1-x$)	(0,0)	(1,2)

Player 1's Expected Utility:

$$\begin{aligned} U_1(x,y) &= x[2y+0] + (1-x)[0+1(1-y)] \\ &= 2xy + 1 - y - x + xy = x(3y-1) + (1-y) \end{aligned}$$

To maximize this, Player 1 looks at the coefficient of x, which is $(3y-1)$.

- If $3y-1>0 \implies y>1/3$, P1 wants x to be as large as possible ($x=1$).
- If $3y-1<0 \implies y<1/3$, P1 wants x to be as small as possible ($x=0$).
- If $3y-1=0 \implies y=1/3$, P1 is indifferent, any $x \in [0,1]$ is a best response.

Player 2's Expected Utility:

$$\begin{aligned} U_2(x,y) &= y[1x+0] + (1-y)[0+2(1-x)] \\ &= xy + 2 - 2x - 2y + 2xy = y(3x-2) + 2 - 2x \end{aligned}$$

Looking at the coefficient of y, which is $(3x-2)$:

- If $x>2/3$, P2 wants $y=1$.
- If $x<2/3$, P2 wants $y=0$.
- If $x=2/3$, P2 is indifferent, any $y \in [0,1]$ is a best response.

If we graph these two best response correspondences, they intersect at three points: 1. $(x=0, y=0) \implies$ Pure NE: (B,R) 2. $(x=1, y=1) \implies$ Pure NE: (T,L) 3. $(x=2/3, y=1/3) \implies$ **Mixed NE:** P1 plays T with prob 2/3, P2 plays L with prob 1/3.

3 The Indifference Principle

Notice in the previous example that in the mixed equilibrium, each player makes the *other* player indifferent between their pure strategies. This is a fundamental property of all mixed strategy Nash Equilibria.

Theorem: Support of a Mixed Strategy

Suppose σ^* is a Nash Equilibrium. Suppose for Player i, the mixed strategy σ_i^* assigns strictly positive probability to two pure strategies, s_i^1 and s_i^2 (i.e., they are in the **support** of σ_i^*).
Then, the expected utility of playing either pure strategy against the opponents' equilibrium profile σ_{-i}^* must be exactly the same:

$$U_i(s_i^1, \sigma_{-i}^*) = U_i(s_i^2, \sigma_{-i}^*)$$

Note

Corollary: For any Nash Equilibrium σ^*, the expected utility $U_i(\sigma^*)$ is exactly equal to the expected utility of **any** pure strategy $s_i \in S_i$ that is played with strictly positive probability (i.e., $\sigma_i^*(s_i) > 0$).

If one pure strategy yielded a strictly higher expected payoff than another, the player would just shift all their probability to the better one, contradicting that σ^* is an equilibrium!